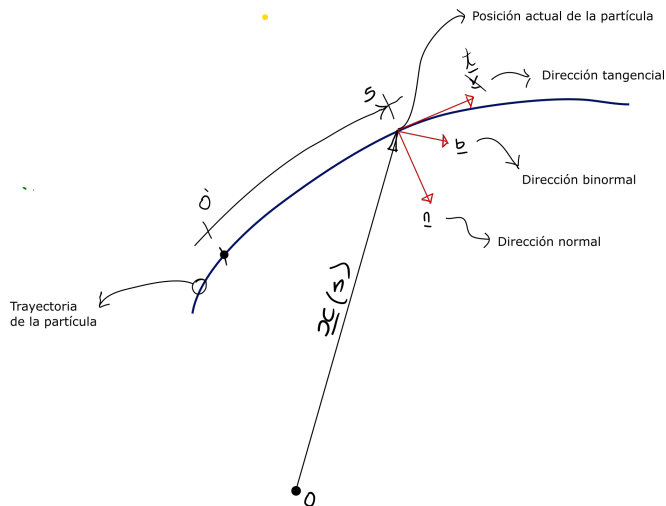


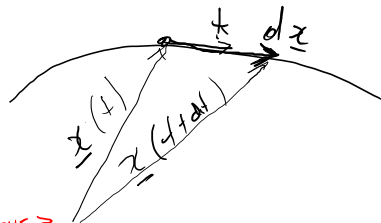


$$\left. \begin{array}{l} \underline{x}(t) \\ \underline{x}(s) \end{array} \right\} \underline{x}(t) = \underline{x}(s(t))$$

• $s(t)$: es una medida de longitud de curva desde O'

• $\frac{ds(t)}{dt} = v > 0$ } ①

• $\underline{t} = \frac{d\underline{x}(s)}{ds}$ } ②



$\|\underline{t}\| = 1$

• $\underline{n} = \frac{d\underline{t}}{ds}$ } ③

curvatura

$\frac{1}{K} \frac{d\underline{t}}{ds}$

$\frac{\Delta \underline{t}}{\Delta s}$

$\Rightarrow \|\underline{t}(s)\| = 1$

$\|\underline{t}(s+ds)\| = 1$

$$(\| \underline{t} \| = \underline{t} \cdot \underline{t} = 1$$

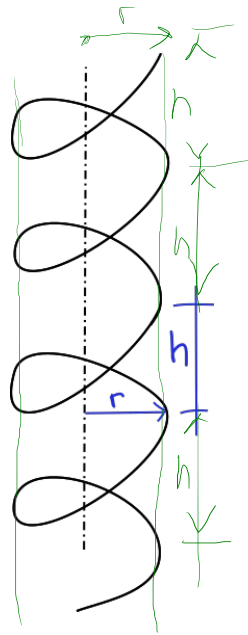
$$\frac{d(\underline{t} \cdot \underline{t})}{ds} = \frac{d\underline{t}}{ds} \cdot \underline{t} + \underline{t} \cdot \frac{d\underline{t}}{ds} = 2 \underline{t} \cdot \frac{d\underline{t}}{ds} = \left. \vphantom{\frac{d(\underline{t} \cdot \underline{t})}{ds}} \right\} \rightarrow$$

$$\frac{d(1)}{ds} = 0$$

$$\boxed{\underline{t} \cdot \frac{d\underline{t}}{ds} = 0}$$

$$k = \frac{\Gamma}{\Gamma^2 + \left(\frac{h}{2\pi}\right)^2}$$

4



$$\underline{x}(s(T))$$

Velocidad de la partícula

$$\underline{v} = \frac{d \underline{x}(s(T))}{dT} = \frac{d \underline{x}(s)}{ds} \cdot \frac{ds}{dT} = v \underline{t}$$

① $\underline{t}$ ② $v = \|\underline{v}\|$

$$\underline{a} = \frac{d \underline{v}(s(T))}{dT} = \frac{dv}{dT} \cdot \underline{t}(s) + v \cdot \frac{d \underline{t}(s)}{ds} \cdot \frac{ds}{dT}$$

③ $k \underline{n}(s)$

$$\underline{a} = \frac{dv}{dT} \cdot \underline{t}(s) + v^2 k \underline{n}(s)$$

$$\underline{a} = \underbrace{\frac{dv}{dt}}_{a_T} \cdot \underline{t}(s) + v^2 \kappa \cdot \underline{n}(s)$$

$$\underline{a} = a_T \underline{t}(s) + v^2 \kappa \cdot \underline{n}(s)$$

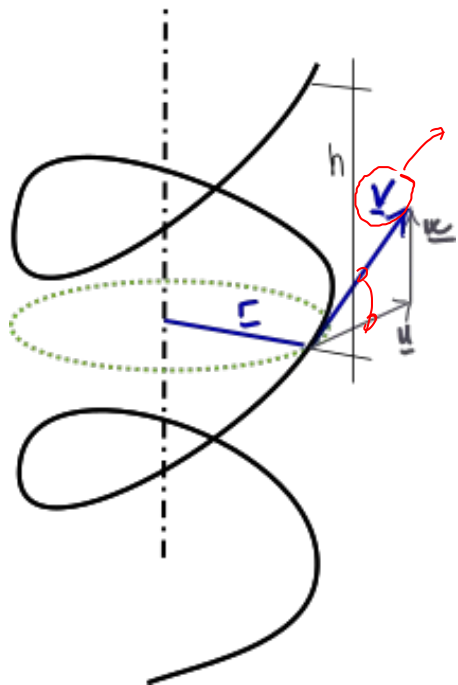
Movimiento de la partícula $\hat{=}$ lo largo de una hélice de paso constante y velocidad constante

- Velocidad constante (v no varía en el tiempo) $\Rightarrow$

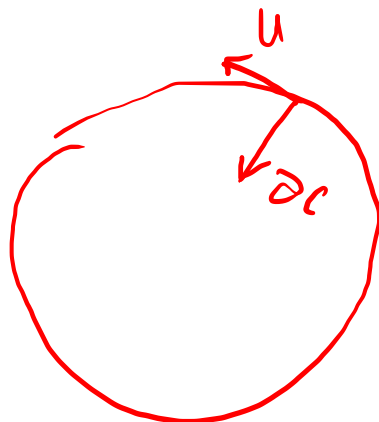
$$a_T = 0$$

Unonda (4) $\sim a_T = 0$

$$\underline{a} = r^2 \cdot \frac{\Gamma}{r^2 + \left(\frac{h}{2\pi}\right)^2}$$



Velocidad
uniforme



$$v_c = \frac{u^2}{r}$$

$$u = \|\underline{u}\|$$

$$\omega = \|\underline{\omega}\|$$

$$v = \|\underline{v}\|$$

son constantes

Tiempo de una vuelta:

$$T_v = \frac{2\pi r}{u}$$

longitud de la circunf.

$$h = \omega \cdot T_v = \omega \cdot \frac{2\pi r}{u} \Rightarrow \omega = \frac{h}{2\pi r} u$$

$$r^2 = w^2 + u^2 \Rightarrow v^2 = \left(\frac{h}{2\pi r}\right)^2 u^2 + u^2$$

$$v^2 = \left(\frac{h}{2\pi}\right)^2 + r^2 \left(\frac{u}{r}\right)^2$$

$$\underline{a} = v^2 \cdot \frac{r}{r^2 + \left(\frac{h}{2\pi}\right)^2}$$

$\Rightarrow$

$$\underline{a} = \frac{u^2}{r} \underline{n(s)}$$